

RESEARCH ARTICLE

Machine Learning-Based Detection of Anomalies, Intrusions, and Threats in Industrial Control Systems

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ABSTRACT Industrial Control Systems (ICS) are critical to the efficient operation of essential sectors such as manufacturing, energy, and water management. However, their increasing integration with IT systems exposes them to sophisticated cyberattacks, particularly lateral attacks targeting Programmable Logic Controllers (PLCs). Advanced preventive measures are necessary because, despite their significance, many ICS continue to rely on outdated technologies with few security features. This paper proposes a machine learning (ML)-based approach to anomaly detection in ICS communication networks, focusing on techniques such as 1D Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Support Vector Machines (SVMs), and Isolation Forest (iForest) algorithms. We generated a dataset by capturing both normal and manipulated ICS communication patterns, including TCP/IP traffic. Simulated lateral attacks provided realistic data for training and testing the ML models. The results demonstrate that the 1D CNN model achieved the highest accuracy (0.92) and F1 score (0.91) with minimal processing time, making it ideal for real-time intrusion detection. This research highlights the potential of ML techniques to fortify ICS cybersecurity and lays the groundwork for future advancements in critical infrastructure resilience.

INDEX TERMS Anomaly detection, intrusion detection systems, machine learning, threat detection, programmable logic controllers.

I. INTRODUCTION

Industrial control systems (ICS) form the backbone of critical infrastructures across multiple sectors. Traditionally isolated, these systems are now increasingly interconnected, exposing them to cyber threats that exploit outdated protocols and hardware. Despite the critical nature of ICS, existing cybersecurity solutions often lack the sophistication required to address modern threats. Machine Learning (ML) techniques provide a promising approach for detecting these threats through advanced anomaly detection systems, enabling real-time identification of intrusions and irregulari-

ties. This paper explores ML-based methods to address this challenge, focusing on their application to programmable logic controllers (PLCs).

The use of ML algorithms allows for the development of robust intrusion detection systems (IDS) that can identify anomalous behaviour and potential security breaches in real time. One approach is to leverage the power of the iForest, a machine-learning algorithm that excels at anomaly detection by isolating normal instances from outliers [1]. Another method involves employing SVMs, a popular supervised learning algorithm known for its ability to handle high-dimensional data and imbalanced sample sets [2], [3]. Additionally, 1D convolutional neural networks (CNNs) have proven effective in various domains and can capture complex

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patterns in sequential data, making them suitable for detecting anomalies in industrial control systems [4].

The aim of this paper is to explore the applicability and effectiveness of these ML methods, including isolated forest, support vector machine, and 1D CNNs, for threat detection, intrusion identification, and anomaly detection in ICS. By utilizing simulated training data divided into packets and labelled with classes (normal and modified), the paper aims to provide insights into the performance and accuracy of these methods. The findings of this research are expected to contribute to the development of robust security measures for industrial control systems, enhancing the protection of critical infrastructures against cyber threats and ensuring the reliability and safety of their operations.

The remainder of this paper is organized as follows. Section II provides the theoretical background, including an overview of PLC vulnerabilities and prior work on ICS anomaly detection. Section III describes the methodology, including the attack simulation and data collection process. The results of the experiments are presented in Section IV, followed by a discussion of their implications in Section V.

II. THEORETICAL BACKGROUND

ICS and their components, particularly PLC's have undergone significant advancements to support industrial automation. However, their increasing integration with modern IT systems has exposed them to new vulnerabilities. This section explores the role of PLCs in industrial automation, highlights their vulnerabilities, and reviews existing research in anomaly detection for ICS.

A. PLCs IN THE IT INFRASTRUCTURE

PLCs are fundamental to ICS, serving as controllers that execute automated processes based on pre-programmed instructions. They are essential for enabling production flexibility, improving efficiency, and ensuring reliability in industrial operations. Recent trends in PLC development have focused on:

- Scalability and Diagnostics: Enhancing the ability to monitor and control complex industrial processes.
- IoT Integration: Connecting devices and sensors to support seamless communication and predictive maintenance.
- Cloud Connectivity: Enabling remote access to data and control mechanisms, improving operational visibility.
- Virtual Commissioning: Simulating and testing control programs before deployment to reduce risks and downtime [5].

Despite their pivotal role, the global PLC market faces significant security challenges. For example, the absence of robust encryption or authentication mechanisms in legacy PLC systems has led to an increased risk of cyberattacks. As the market is projected to grow to USD 18.3 billion by 2030, securing these systems has become a critical priority. The impact of the control systems and industrial automation

is huge. The market was estimated at ca 170 billion USD (2022) and the expectations in 2023 are reaching 188 billion USD [6].

The global PLC market was valued at USD 11,6 Bn in 2021 and the projected value in next years, e.g., the projection in USD 18.3 Bn by 2030 [7].

Projected size of the global programmable logic controller (PLC) market between 2020 and 2026: (in billion U.S. dollars) [8], shown in Fig.1.

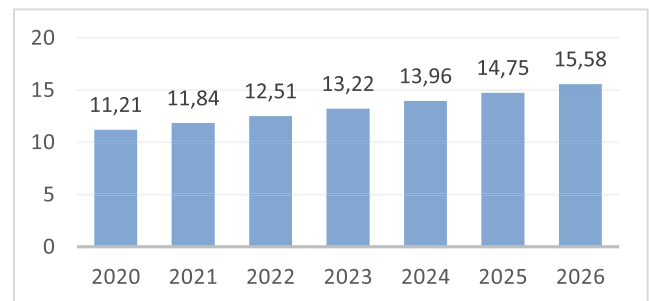


FIGURE 1. Market size - PLC [8].

For a better illustration of the PLC market, the compound annual growth rate (CAGR) is used as the parameter showing the average rate of revenues. The estimation of the PLC growth is at a CAGR of 4.1% between 2022 and 2027 and the size of the market is forecast to increase by USD 3,289.73 million [7].

B. CYBERSECURITY AND THREAT LANDSCAPE

In a rapidly evolving security landscape of Europe and beyond, the need for innovative solutions and cooperation is necessary. The growing presence of renewable energy resources, physical electric infrastructure, smart grids, and Industry 4.0 technologies, including control systems, SCADA, or billions of IoT devices, introduces new security challenges. Cyberattacks on critical energy infrastructure can have significant economic consequences, as seen in the case of the U.S. power grid, released by Lloyd's of London, notes the scenario's impact on the U.S. economy is estimated at more than 240 billion dollars.

Generally, critical infrastructure assets could be destroyed or degraded by both natural and human-initiated threats. Any disruption, whether stemming from a terrorist attack, a natural disaster, or man-made damage is likely to create a cascading effect on any country's energy system that is both complex and interconnected [15]. Symantec (also known as Norton) observed during one year of monitoring an average of 74 targeted attacks per day globally, of which nine attacks (16.3% of all attacks) per day were targeted towards the energy sector [16]. Cyber-attacks present a new category of security threats with little expertise or experience in critical infrastructures or in industry at all.

As mentioned by the Commissioner for Humanitarian Aid and Crisis Management Stylianides in 2014 [24] in his

confirmation hearing to the European Parliament, 1 Euro invested in resilience saves 7 Euro in emergency.

Attacks on ICS can range from disabling security apparatus to inflicting direct physical damage, as highlighted by reference [25]. The security of such systems has been a growing concern, with instances of simple code injection proven by researchers in references [26], [27], [28], and more intricate attacks like Stuxnet, underscoring the urgency for advanced protective measures [19], [29]. Siemens has been incorporating cryptographic mechanisms in newer products, such as the S7-1500 family, to overcome detected vulnerabilities [30]. However, vulnerabilities persist, with Rockwell Automation PLC experiencing a notable one, marked as CVE-2022-1161, allowing undetected internal code modifications [31]. Mitsubishi PLCs are not exempt, with vulnerabilities like CVE-2022-29831, CVE-2022-29832, and CVE-2022-29833 enabling attackers to compromise connected safety CPU modules by accessing information from GX Works3 project files [32].

Finding more and more vulnerabilities shows how important it is to keep improving security to protect against many different cyber threats. Combining new security features, regular updates, and better awareness of security is key to making these systems stronger against cyberattacks. The combined efforts of developers, security experts, and those in the industry are vital in dealing with the complex security challenges and making these systems more reliable.

C. PLC VULNERABILITIES

PLCs, while critical to industrial automation, are increasingly targeted by cyberattacks as they integrate with modern IT systems. Key vulnerabilities stem from their legacy designs, which often lack robust authentication, encryption, and update mechanisms. For example, outdated firmware and protocols leave PLCs susceptible to attacks such as unauthorized data manipulation, denial-of-service (DoS) attacks, and lateral assaults. Addressing these vulnerabilities requires proactive measures, including firmware updates, advanced intrusion detection systems, and adherence to cybersecurity best practices.

The security of PLC systems is a critical consideration, especially in industrial and critical infrastructure settings, where they are used to control vital operations. Common types of vulnerabilities associated with PLCs:

- Outdated Hardware and Software: Many PLCs lack updates against known vulnerabilities, leaving them susceptible to exploits [9].
- Weak Authentication and Access Control: Insufficient mechanisms allow unauthorized users to modify PLC configurations or commands [10].
- Lack of Encryption: Unencrypted communication channels expose data to interception and manipulation [11].
- Susceptibility to Malware and Ransomware: Attacks such as Triton and Stuxnet have demonstrated the capability to disrupt PLC operations, causing significant physical and financial damage [12], [13].

These vulnerabilities pose risks not only to industrial processes but also to critical infrastructure sectors like energy, water, and transportation.

To enhance the security of PLC systems, measures such as firmware updates, strong authentication and access control, data communication encryption, network monitoring, and threat detection are crucial. Industrial organizations should also follow best practices for cybersecurity and have incident response plans in place in case of an attack. Ensuring the security of PLC systems is a critical aspect of industrial automation and process control [14].

D. MACHINE LEARNING ALGORITHMS

iForest is a type of unsupervised ML method used to discover anomalies. With recursive partitioning, it isolates anomalies and is useful for high-dimensional data. The technique isolates instances that require fewer partitions than typical instances and it has a unique ability to discover anomalies [33]. Because of its effectiveness and capacity to manage both numerical and categorical data types, iForest is a great option for promptly spotting odd communication patterns that could be signs of a cyberattack or other system issues.

Tasks involving anomaly detection and image processing frequently make use of CNNs. Because CNNs can automatically learn hierarchical feature representations from data, they are a powerful tool for identifying spatial anomalies in intricate industrial processes. Convolutional filters are applied to the input data by CNNs, which helps to find patterns or anomalies by capturing significant features at various levels [34]. CNNs are useful for identifying abnormalities such as abrupt changes in communication frequency or unexpected data flows inside the network because communication patterns in ICS frequently have temporal-spatial correlations that they can learn.

Recurrent neural networks (RNNs) of the LSTM type are well suited for temporal anomaly detection in ICS, because of their superior ability to handle sequential input. Because of its memory cells' ability to store data for extended periods of time, LSTM networks can identify temporal dependencies in time-series data. Through effective learning from past communication sequences, LSTM can eventually detect patterns that deviate from the norm and could point to cyber threats, latency problems, or faults in the system [35]. Because of this, LSTM is perfect for identifying irregularities in industrial processes that show temporal fluctuations.

SVM is a supervised learning algorithm that is widely used for classification and regression tasks. SVM works by finding the optimal hyperplane that separates data points into different classes, making it suitable for anomaly detection in industrial control systems where classification of normal and anomalous data is essential. SVM can effectively handle high-dimensional data and has been shown to perform well in identifying anomalies [36]. When the ICS' data contains some identified instances of both normal and anomalous communication behaviour, SVM can be helpful. Its capacity to draw a distinct decision boundary facilitates the detection

of communication traffic anomalies that may indicate vulnerabilities or assaults.

Each above-mentioned method targets a different area of anomaly detection. CNN is good at capturing spatial patterns, LSTM is good at focusing on temporal relationships, SVM is good at classification, and iForest is good at handling high-dimensional data. Using these algorithms enables thorough identification of different kinds of anomalies in ICS transmission.

Traditional IDS often rely on signature-based techniques, which are limited in detecting novel or zero-day attacks. In contrast, anomaly detection systems leverage ML to identify deviations from normal behavior. Recent research has focused on the use of ML and artificial intelligence (AI) for anomaly detection in automation environments, notably ICS and the Industrial Internet of Things (IIoT). Several strategies have been developed, each with the goal of addressing certain issues inherent in these systems, including real-time processing, high accuracy, and adaptation to unexpected threats.

However, it is important to note that some tools, like Snort, primarily represent the signature-based detection paradigm, despite the existence of third-party add-ons that incorporate AI/ML components. Snort, developed by Cisco Systems, is one of the most widely used open-source intrusion detection and prevention systems (IDS/IPS/IDPS) in the world [37]. It can be run in one of three modes: packet sniffer mode, packet logger mode, and network intrusion detection and prevention system mode. While some ML preprocessors can be integrated into Snort, these are mostly third-party additions and are not native features of the system. Thus, Snort remains fundamentally a signature-based IDS.

The effectiveness and applicability of various IDS have been analysed extensively in literature. In Table 1 below, we summarize key studies on IDS tools, including their objectives, metrics, strengths, and limitations.

Traditional IDS rely heavily on signature-based techniques, which are successful against known threats but have limited ability to detect novel or zero-day assaults. Anomaly detection systems that use ML offer a considerable advantage by spotting deviations from regular behavioral patterns. Recent breakthroughs have investigated the use of ML and AI for anomaly detection in automation environments such as ICS and the IIoT. These initiatives are intended to address intrinsic issues such as real-time processing, high precision, and adaptability to unexpected threats.

Snort, among classic IDS technologies, is a prominent example of the signature-based detection paradigm. Snort, developed by Cisco Systems, is extensively used for its functionality and substantial community support. It operates in three modes: packet sniffer, packet logger, and network intrusion detection/prevention system, allowing for flexible deployment options. However, its reliance on predetermined signatures limits its usefulness against new threats. While AI/ML preprocessors can be added to Snort via third-party add-ons, these functionalities are not native, hence Snort remains primarily a signature-based IDS.

TABLE 1. Comparative analysis of Intrusion Detection Systems (IDS).

Tool	Objective	Metrics	Strengths	Limitations
Snort	Signature-based IDS with optional AI/ML preprocessing	Detection accuracy, processing speed	Widely used, extensive community support [37]	Primarily signature-based, AI/ML features are third-party add-ons Complex configuration, not straightforward for basic IDS/IPS needs
Zeek (Bro)	Network analysis and security monitoring	Network visibility, analysis depth	Detailed insights into network activity [39]	Can be resource-intensive, complex setup in large networks
Suricata	High-performance, scalable network-based IDS/IPS	Scalability, throughput, detection accuracy	Efficient multi-threading, handles high traffic volumes [39]	

The above Table 1 provides a comparison of IDS tools, highlighting their objectives, capabilities, and limitations. Snort, Zeek, and Suricata are evaluated using parameters such as detection accuracy, processing speed, scalability, and flexibility. Snort’s signature-based method remains a drawback, despite its ease of usage and widespread acceptance. Zeek excels at delivering detailed network analysis, although it can be unnecessarily complicated for basic requirements. Suricata’s multi-threaded architecture makes it exceptionally efficient at managing enormous traffic volumes, but it is resource-intensive, causing issues in large-scale implementations.

To address the limitations of classical IDS, researchers have increasingly concentrated on machine learning-based anomaly detection systems. Studies such as those in [38] and [39] emphasize the shift to data-centric anomaly detection, meaning the importance of data quality and feature engineering in strengthening cyber resilience. Reference [38] compares supervised and unsupervised ML techniques for anomaly detection, focusing on the challenges of feature extraction and selection in the heterogeneous ICS context.

Behavioral intrusion detection, as discussed in [1], emphasizes the need of strong feature selection and ICS-specific metrics. This study categorizes existing techniques and identifies flaws such as the absence of adaptive learning mechanisms that can change with dynamic ICS operations. Reference [20] dives into the complexity of installing ML detectors in big industrial facilities, emphasizing the importance of scalable, real-time learning models for efficiently processing changing data streams.

Hybrid detection methods that integrate statistical and machine learning methodologies are described in [2] and [4]. Reference [2] describes a two-stage model that combines ARIMA for time-series forecasting and one-class SVM for anomaly detection, improving detection accuracy. Similarly, [4] presents a system that combines 1D CNN, autoencoders,

and PCA to reduce false positives while increasing detection precision in ICS.

Deep learning models, such as CNNs and LSTMs, have shown particularly promising results when dealing with complex ICS data. References [42] through [44] show the potential of hybrid CNN-LSTM architectures for anomaly detection, with good accuracy in benchmarks such as CICIDS2017. These models accurately capture spatial and temporal patterns in ICS data. Furthermore, as indicated in [45], iForest has demonstrated potential in anomaly detection tasks, particularly in cases when labeled data is limited.

To enhance or supplement these prior attempts, we introduce datasets and concentrate on a distinct side of intrusion detection that was not covered by the earlier publications. We are certain that our strategy provides a novel viewpoint and enhanced outcomes for our study issue. The following table summarizes the differences between our paper and other works.

TABLE 2. Summarization of differences between our paper and other works.

Attribute	Previous papers	Our paper
Intrusion Detection Methods	Traditional ML (e.g., SVM, kNN, Decision Trees) or unsupervised learning	Precision tailored ML (one class SVM, 1D CNN)
Evaluation metrics	High detection rates (accuracy, precision, recall)	Enhanced metrics (F1 score) for real-time detection
Data Sources	Static datasets (e.g., KDD Cup, NSL-KDD)	Real-time ICS communication data
Focus of Research	Generic intrusion detection in IT networks, anomaly detection in spatial data	Targeted anomaly detection in ICS communication
Adaptability / Scalability	Limited to specific anomaly types	Adaptability to various intrusion types in ICS networks

E. MOTIVATION

The integration of ICS with contemporary IT networks has significantly increased the need for sophisticated security measures to protect critical infrastructure. PLCs and other ICS components are indispensable for industrial operations, but their inherent vulnerabilities make them prime targets for cyberattacks. Such attacks can have catastrophic consequences, as demonstrated by notable incidents over the past decade.

In 2014, a cyber-attack targeted energy companies, infecting 250 organizations across the USA and Western Europe [17]. The following year, a major attack on electricity operators in Ukraine disconnected 30 substations, leaving eight provinces without electricity for several hours. This incident impacted more than 200,000 people and caused physical damage to ICS infrastructure [18]. Similarly,

in 2016, another cyber-attack on Kyiv's power grid caused widespread outages and revealed the use of malware specifically designed to compromise ICS and SCADA systems [19].

The 2017 Triton malware attack on a petrochemical plant in Saudi Arabia further illustrated the destructive potential of ICS-targeted cyber threats. By crippling the plant's safety systems, this attack underscored how adversaries could directly endanger human lives [20]. In 2019, an unnamed virus partially shut down the factory operations of the eye-glass lens manufacturer Hoya in Thailand, causing significant operational disruptions [10].

More recently, the ransomware attack on Colonial Pipeline in 2021 disrupted the fuel supply across the Eastern United States, leading to widespread fuel shortages and economic ramifications [21]. In 2022, a cyberattack on the wind turbine manufacturer Nordex SE in Germany disabled remote access to wind turbines, severely affecting their operation and maintenance capabilities [22]. Finally, in 2023, a ransomware attack on ABB, a global leader in industrial automation, highlighted the ongoing vulnerabilities in critical industrial infrastructure [23].

These incidents collectively demonstrate how adversaries exploit vulnerabilities in ICS to inflict financial, operational, and physical harm. A particularly concerning attack vector is lateral movement within networks, where intrusions spread by exploiting flaws in protocols like TCP/IP and UDP. Such attacks frequently bypass conventional perimeter defenses and pose a significant threat to the integrity of industrial systems. Addressing these challenges requires advanced anomaly detection mechanisms capable of identifying unusual communication patterns that signal potential threats.

Machine learning (ML) provides a promising solution for this critical issue by enabling real-time monitoring and adaptive anomaly detection in ICS communications. This study evaluates the effectiveness of various ML models, including Isolation Forest (iForest), Support Vector Machines (SVMs), Long Short-Term Memory (LSTM) networks, and 1D Convolutional Neural Networks (CNNs). These models are assessed to determine their suitability for detecting sophisticated lateral attacks and ensuring the security of ICS networks.

III. METHODOLOGY

In this section, the proposed ML anomaly detection algorithms are described, followed by a description of data collection from industrial network.

The methodology consists of several stages: problem identification, attack mitigation strategy, algorithm selection, data collection, and experimental setup. Each stage is designed to ensure comprehensive evaluation and effective anomaly detection.

A. PROBLEM IDENTIFICATION

First, we identify the types of attacks that may occur on the ICS and the common ways to mitigate them. Lateral attacks pose a unique challenge within ICS networks due to their

ability to mimic normal communication patterns. This paper identifies key types of anomalies indicative of such attacks:

- Unauthorized Data Manipulation: Injection of rogue commands or altered data packets.
- Communication Floods: Overwhelming devices with excessive traffic.
- Protocol Exploitation: Misuse of TCP/IP or UDP protocols to gain unauthorized access.

The primary objective is to detect these anomalies by analyzing communication patterns between PLCs.

B. ATTACK MITIGATION STRATEGY

The chosen approach to mitigate lateral attacks is through anomaly detection, where anomalous communication patterns are identified as potential security threats. Anomaly detection is a critical process across many disciplines, allowing for the discovery of unusual occurrences or behaviors. By learning patterns from large datasets and successfully detecting abnormalities, ML approaches have emerged as strong tools for automating anomaly identification.

The approach emphasizes:

- Pattern Recognition: Detecting irregularities in communication frequency, packet structure, or data payloads.
- Real-Time Processing: Ensuring anomalies are identified promptly to minimize operational disruptions.

C. ALGORITHM SELECTION

We selected several ML techniques based on their performance in existing studies for ICS anomaly detection. The algorithms considered include the iForest, CNNs, SVMs, and LSTM networks. The iForest is a tree-based anomaly detection approach that builds an ensemble of random decision trees. Because of their shorter average path lengths inside the tree structure, each tree separates cases that are likely to be anomalies. iForest has proven to be successful in detecting abnormalities in ICS datasets [1].

CNNs, which are well-known for their performance in image and signal processing applications, have also been used in ICS anomaly detection. CNNs can discover aberrant activities by evaluating time-series sensor data or photos taken within the ICS environment and extracting spatial and temporal patterns. The use of CNNs in ICS anomaly detection has yielded encouraging results [40].

SVMs have been widely employed in anomaly detection due to their ability to capture complicated decision boundaries. SVMs may be used in ICS to identify cases as normal or abnormal based on extracted characteristics. In ICS anomaly detection research, SVMs have proven their adaptability and performance [41].

LSTM networks are a form of RNNs developed to solve the vanishing gradient problem and capture long-term relationships in sequential data. LSTM networks, unlike standard RNNs, include neurons that can store information for long periods of time, allowing them to retain and transmit useful

information while filtering out unnecessary or duplicate information. LSTM networks have also been used as a deep learning approach in the context of ICS intrusion detection to improve the accuracy and efficacy of IDS. These solutions are designed to detect and prevent security breaches or hostile activity in industrial networks.

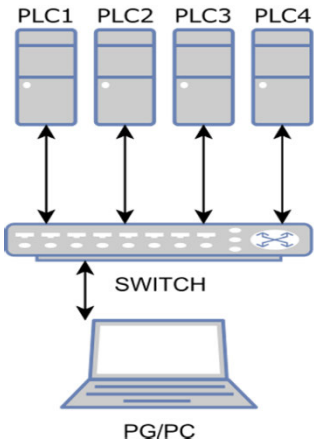


FIGURE 2. Used network topology.

D. DATA COLLECTION SETUP

For data collection, we proposed a system that is hardware-neutral, meaning it is not limited to Siemens PLCs; it can also be used with PLCs from other manufacturers. However, for our experimental setup, as seen in Figure 2, we used four independent S7-300 PLCs connected via a NETGEAR GS305E 5-port manageable switch. The choice of S7-300 PLCs was motivated by the usage of a PLC laboratory at our institution. Based on our experience, this type of PLCs is still widely used in industry up to this day.

The switch was configured to clone packets to port 5, to which the Simatic field PG was connected. A simple packet sniffer was running on the PG, storing captured data into separate files. Physical connection was realized using UTP Cat. 5E Profinet in cable. For configuration purposes, we used Simatic Manager and its NetPro tool, where the communication was set as shown in Table 3

TABLE 3. Simplified communication configuration table.

Parameter	Value	Type	Active
Number of estimators	10, 25, 50, 100, 200, 500, 1000	S7 connection	Yes
Max samples	0.5, 0.75, 1	S7 connection	Yes
Contamination	0.01, 0.05, 0.1	S7 connection	Yes

E. EXPERIMENTAL SETUP

To collect data, packet sniffing was carried out in three distinct instances:

```

Network 1: Title:
      AN      M      0.0
      =      M      0.0

Network 2: Title:
      CALL    "GET" , DB1
      REQ     :=M0.0
      ID      :=W#16#1
      NDR     :=#nдр
      ERROR   :=#err
      STATUS  :=#sta
      ADDR_1 :=DB10.DBD0
      RD_1   :=DB10.DBD0

Network 3: Title:
      CALL    "PUT" , DB2
      REQ     :=M0.0
      ID      :=W#16#3
      DONE    :=#done
      ERROR   :=#err
      STATUS  :=#sta
      ADDR_1 :=DB10.DBD0
      RD_1   :=DB10.DBD0

```

FIGURE 3. Structure text diagram used instructions.

- During the foundational communication among the configured PLCs.
- Basic communication between the configured PLCs occurred with Simatic Manager set to “Online,” allowing for the reading of the actual state from the PLCs.
- While uploading the program (with an altered DB) amidst the pre-configured communication among the PLCs.

An example of structure of the PLC connection with used instructions is shown in Figure 3.

Two networks were made for the PLC structure using the “GET” and “PUT” instructions for the PLCs to communicate with one another. In an ICS, these instructions are essential for data exchange, guaranteeing effective and seamless communication between linked equipment.

Data is be read from a PLC or its partner PLC using the “GET” instruction, which is shown in Network 2. The REQ input, which must be set to true to start the procedure, is what triggers the instruction. The ID parameter establishes the communication path and identifies the connection. The signal NDR indicates that new data is ready after a successful data transfer, while the ERROR and STATUS outputs reveal any circumstances or problems that occurred during execution. The ADDR_1 parameter, which indicates where the source data is in the remote device, and the RD_1 parameter, which establishes the local memory address where the received data is placed, define the data transfer itself.

In Network 2, the “PUT” instruction is used to write data to a partnering PLC. The process is started when the REQ input is triggered, same as the GET operation. The data is correctly routed to the intended communication channel thanks to the connection ID. The DONE signal verifies the data transmission once the PUT operation is successful, while the ERROR and STATUS outputs offer diagnostics if there

are communication problems. ADDR_1 and RD_1 define the data source and destination addresses, which guarantee that the sent values are correctly placed in the local and distant memory areas.

The communication process is largely controlled by the ports and parameters that are utilized in both networks, including REQ, ID, NDR, DONE, ERROR, and STATUS. These parameters ensure the system’s robustness and dependability by starting and finishing the processes as well as providing the required diagnostic data. When working in complex network environments, the system can differentiate between several channels thanks to the connection IDs (W#16#1 for GET and W#16#3 for PUT), which uniquely identify the communication paths.

The addresses that ADDR_1 and RD_1 specify are essential for figuring out precisely where data is to be read or stored in the PLC’s memory. As an illustration of a clear and efficient data transmission, the given structure uses DB10.DBD0 as both the source and destination address.

For industrial applications that demand quick updates and reactions, this structure makes it possible for PLCs to communicate in real time. In order to facilitate speedy diagnostics, it also includes error handling methods. ERROR and STATUS outputs promptly signal communication failures or irregularities. By enabling targeted data transfers to memory blocks, the GET and PUT instructions provide flexibility, and their well-defined input and output ports make it easier to integrate them into pre-existing PLC applications.

F. ATTACK SCENARIOS

An operating ICS environment with regular instructions being sent between PLCs was used to record normal traffic. On the other hand, malicious or unusual traffic was purposefully added to produce a varied dataset for model training. For the anomalous traffic, packets were intentionally modified to simulate realistic intrusion attempts. Attack scripts introduced the following anomalies in packet contents and structure: altered IP headers, manipulated payload contents, and inserted commands to simulate unauthorized access attempts. The ML models were able to learn and identify the distinctions between potentially harmful and benign traffic patterns thanks to this setup

To evaluate the proposed ML-based anomaly detection system, a series of realistic attack scenarios were designed and implemented. These scenarios simulate common vulnerabilities and exploits within ICS, focusing on attacks targeting PLCs through lateral communication channels.

Simulating and testing various attack scenarios in ICS environments is essential for thorough anomaly detection and intrusion analysis. Both malicious and benign packet transmissions over TCP protocols were included in the attack scenarios, which covered both standard PLC traffic flows and altered patterns intended to resemble possible real-world attack traffic. We simulated common attacks including packet replay attacks, unauthorized data modification, and command injection.

The dataset was generated using a simulated PLC environment with four independent PLCs communicating over a managed network switch. The data collection process included the following steps:

- **Baseline Traffic:** Normal communication traffic was captured during standard PLC operations.
- **Simulated Attack Traffic:** Anomalous traffic was introduced by:
 - Injecting rogue TCP/IP packets.
 - Modifying data payloads to simulate unauthorized commands.
 - Creating communication floods to disrupt normal operations.

Because of its reputation for dependability, TCP was selected for attacks like data manipulation and unauthorized command injection that call for extended connections. The attackers sent rogue packets with altered instructions after establishing a three-way handshake. To ensure the dataset was balanced, approximately 70% of the traffic was normal, and 30% consisted of attack scenarios. Packet metadata, protocol information, and payload content were recorded for analysis. Certain IP packet characteristics were altered to mimic hostile attacks to increase the dataset’s realism. IP addresses of source and destination nodes were changed to imitate attackers trying to get around access control or authentication. This technique is known as source and destination spoofing. Payload modification was used to simulate unauthorized or altered commands included in data payloads. To mimic man-in-the-middle (MITM) attacks, in which packets are intercepted and changed before they reach their destination, some IP header fields, including the time-to-live (TTL) and sequence numbers, were changed. We were able to produce increasingly complex and realistic assault scenarios through these changes, which qualified the dataset for use in training and testing machine learning models for intrusion detection.

G. DATA COLLECTION SETUP

Both raw hexadecimal and binary forms of the payload instructions and metadata (such as source and destination IP addresses, ports, and timestamps) were included in the packet captures. This thorough gathering made sure that the dataset was balanced between attack and regular data and indicative of different forms of traffic.

GET and PUT blocks built into the PLCs’ firmware, set as stated in Table 4, helped PLCs communicate with one another.

TABLE 4. Communication GET/PUT blocks.

Partner 1	Function	Partner 2
PLC 2	GET	PLC 1
	PUT	PLC 3
	GET	PLC 3
	PUT	PLC 4
	GET	PLC 4

Data packets related to communication were mirrored by the network switch to the fifth port, where a sniffer was installed in the PG/PC station to capture the packets. These packets were then stored in hexadecimal format as raw data, which were subsequently used to create the ML model’s training dataset.

The 70-30 ratio of data division allowed us to keep the balance between the classifications quite well. This is significant because a model may perform well by just predicting the majority category in a highly imbalanced dataset, which could lead to deceptive accuracy estimates. We also used metrics like precision, recall, and F1-score to evaluate model performance more accurately to address this problem.

Most of the anomalous traffic comprised of simulated lateral attacks, which were meant to imitate realistic attack vectors. These included illegal data access and manipulation between PLCs. The attacks were created by combining replay assaults — in which malicious behaviour was added to previously recorded regular traffic—with carefully constructed attack scripts that attempt to take over PLCs without authorization or insert fake data. This method made sure that both covert and overt intrusion attempts were recorded in the dataset.

The sniffer used for data acquisition was written in C# using the SharpPcap library, which allows for efficient packet capture. After initializing a list of active devices, it asks the user to choose an adapter from the list of choices. It configures a FileStream to write the captured packets to a designated file upon user selection; it handles file capacity limitations by generating additional files as needed. Error handling made sure that data capture remained reliable even when wrong inputs were present.

To ensure thorough data collecting for this paper, the sniffer recorded all data sent over the network, without any filters. The key features that were retrieved and examined were:

- **Packet Metadata:** Each packet contains its source and destination IP addresses, ports, and timestamps, giving a comprehensive picture of the communication flows between devices.
- **Protocol Information:** Data about protocols (e.g. TCP and UDP) to examine the many kinds of network connections.
- **Payload Content:** Each packet contains raw data that provides insight into the real data that is transmitted between PLCs and other network devices.
- **Traffic Volume and Timing:** Analysis of patterns in network usage by analysis of packet size, frequency, and transmission timing can help spot anomalies.

These features provided a comprehensive understanding of both typical operations and possible security risks. We addressed the potential concerns with class imbalance by carefully building the dataset to ensure that both regular and attack traffic were appropriately represented. To prevent skewing the model’s learning process toward the majority

class (regular traffic), the dataset was specifically created to maintain a 70:30 ratio of normal to attack traffic.

To achieve this balanced representation, we implemented the following steps:

- Data augmentation: To simulate a variety of lateral attacks, including command injections, data manipulation, and illegal access, we added synthetic instances to the attack stream.
- Stratified sampling: To improve the generalizability of the model, we kept the class distribution constant across the test, validation, and training sets.
- Comprehensive metrics: In addition to accuracy, the model’s ability to identify anomalies was verified using precision, recall, F1-score.

The dataset consists of 10000 packets, with a balanced distribution of 70% normal traffic and 30% anomalous traffic.

The data collection process involved recording normal traffic during standard PLC operations and generating anomalous traffic by simulating realistic cyber-attack scenarios. These scenarios included rogue TCP/IP packets, unauthorized command modifications, and communication floods, which were designed to represent potential threats in industrial control system environments.

Each packet in the dataset was captured in its raw hexadecimal and binary formats, providing detailed insights into the communication patterns. Additional metadata, such as source and destination IP addresses, timestamps, and protocol information, was included alongside the payload. These comprehensive details ensure that the dataset is well-suited for training and evaluating machine learning models in detecting anomalies within industrial control system communications.

H. MACHINE LEARNING ALGORITHMS

We employed several ML algorithms. Those include a 1D CNN, an iForest a one-class SVM, and a LSTM network.

Our paper’s ML algorithms function in a serial structure, meaning the layers in each algorithm are connected sequentially one after the other.

The data were organized into two categories, normal and unusual communication. Normal communication means, there are only the instructions running, and the data flow is captured. Unusual communication means that there is a change in the communication. This includes changes in the instructions, connection of an unknown device etc.

The way the layers of the used algorithms are connected in a series is shown in the figures below. Before sending its processed output to the next layer, each algorithm operates independently to carry out its designated task. Working in a serial manner allows the system to fully comprehend the data because each layer of the algorithm concentrates on a different component of the data, improving the overall accuracy and detection capability.

Our first ML algorithm used was a LSTM network. The architecture is shown in Fig. 4.

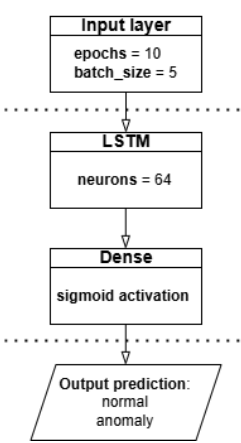


FIGURE 4. LSTM architecture used.

We used a simple LSTM architecture with a dense layer. The activation used was a sigmoid function. Dropout and L2 regularization were used as regularizations as these are highly effective in enhancing the performance and generalizability of LSTM models, especially when working with time-series data in anomaly detection tasks. All the algorithms we used were trained on 10 epochs and with a batch size of 5. The number of neurons, 64, in the LSTM layer was chosen based on the testing results using the grid search method. Other values we experimented with include 2, 8, 16, 32, 128 neurons.

Our next ML algorithm used was iForest. The initial parameters of the forest are summarized in Table 5. Out of those, we chose optimal parameters using grid search and based on the test results of the algorithm.

TABLE 5. Training parameters of the iForest algorithm.

Parameter	Values
Number of estimators	10, 25, 50, 100, 200, 500, 1000
Max samples	0.5, 0.75, 1
Contamination	0.01, 0.05, 0.1

The optimal parameters we used are as follows: number of estimators 100, maximum number of samples 1, and contamination 0,05.

The next algorithm used was a linear SVM. The initial training parameters are summarized in Table 6.

TABLE 6. Training parameters of the linear SVM algorithm.

Parameter	Values
C	10, 25, 50, 100, 200, 500, 1000
Tolerance	0.5, 0.75, 1
Number of iterations	0.01, 0.05, 0.1
Penalty	L1, L2

To find the optimal parameters for our SVM we used a grid search. The optimal parameters were C 0,01, tolerance 5, number of iterations 10000, and L1 penalty.

The last algorithm we employed was a 1D CNN. The architecture is depicted in Fig. 5.

All the above-mentioned ML algorithms were trained in binary and decimal format of the captured communication packets. A total of 10000 packets were used for the training of ML algorithms and 100 packets for testing purposes.

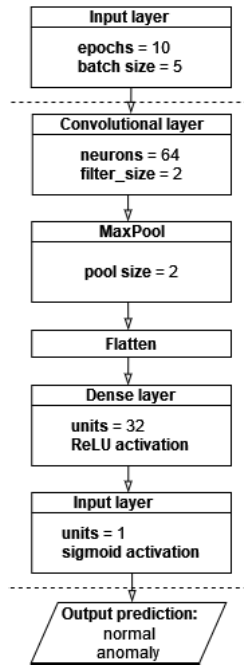


FIGURE 5. CNN architecture used.

IV. RESULTS

The experimental results are shown in Table 7. For testing purposes, we used 25 files containing packets within the anomalous category. We monitored training and testing accuracy as well as the F1 score during the tests. The following table contains the values of the mentioned metrics, and several other metrics.

In terms of accuracy and F1 score, the 1D CNN model surpassed both the initial and traditional CNN models. The training accuracy was 0.91, and the testing accuracy was 0.92, demonstrating the usefulness of the 1D CNN architecture in identifying threats, intrusions, or abnormalities in ICS. In terms of processing time, the 1D CNN model performed admirably. The average processing time per file was about 0.4s making it a viable option for real-time threat detection in industrial control systems where prompt reactions are critical. The reduced processing time not only improves the system’s responsiveness but also optimizes resource usage.

The success of the 1D CNN model opens new opportunities for its use in a variety of industrial contexts. Its capacity to evaluate sequential data makes it ideal for time-series analysis jobs such as anomaly detection in control system communications. Industries that rely on complex control systems, such as manufacturing plants, electricity grids, and water treatment facilities, can benefit from the 1D CNN model’s increased

accuracy and efficiency. In the following table is a summary of the steps of each algorithm and the improvement.

TABLE 7. Testing results of each algorithm.

Algorithm	LSTM	iForest	1D CNN	SVM
Training accuracy	0.89	0.72	0.91	0.51
Testing accuracy	0.75	0.70	0.92	0.52
F1 score	0.77	0.68	0.91	0.51
True positive	18.75	17.5	23	13
False positive	18.75	22.5	5.9	36
True negative	56.25	52.5	69	39
False negative	6.25	7.5	1.9	12
Precision	0.5	0.44	0.79	0.27
Recall	0.75	0.7	0.92	0.52
Testing time [s]	2.5	3	0.79	4.8

The 1D CNN is the ideal option for ICS where precision and recall are crucial since it obtained the highest accuracy and F1 score, proving its exceptional performance in anomaly detection. While the performance levels of the other algorithms (LSTM, iForest, and SVM) varied, none of them were as good as the CNN in terms of accuracy and efficiency.

We have found three similar studies to our research, [51], [52], and [53]. Reference [51] uses a time-series memory augmented autoencoder (TSMAE) for anomaly detection in IoT and [52] uses SimpleNet for image anomaly detection and localization. Those are more specialized, and the AutoML method used for intrusion and anomaly detection in IoT in [53] seeks to minimize human involvement, our approach is more methodically hand-picked and complex, integrating various algorithms that provide comprehensive and adaptable detection adapted to ICS.

TABLE 8. Steps (Data flow) for each algorithm and our improvements.

Algorithm	Steps	Improvements
1D CNN	1. Input Data → 2. Convolution Layer → 3. Pooling → 4. Flattening → 5. Activation → 6. Fully Connected Layer → 7. Output	Enhanced filter sizes, dropout layers, batch normalization
iForest	1. Data Sampling → 2. Tree Construction → 3. Scoring → 4. Thresholding	Feature scaling, optimized contamination fraction
Linear SVM	1. Input Data → 2. Model Training → 3. Decision Function	Kernel selection (RBF), penalization (L1, L2)
LSTM	1. Input Sequence → 2. Memory Cell Operations → 3. Hidden State Propagation → 4. Output	Regularization techniques (L2, dropout)

When it comes to ICS anomaly detection, our work is excellent at striking a balance between efficiency and accuracy. Although SimpleNet is more suited for image data and has a 99.6% AUROC score, it is less directly comparable to our packet-based ICS application. Although it lacks precise metric comparisons, TSMAE may provide competitive performance for time-series data, and the AutoML method puts

automation ahead of raw performance. Compared to the more specialized models (TSMAE and SimpleNet), our work offers a unique blend of flexibility and adaptability targeted to ICS application. While the AutoML technique is flexible, it lacks the depth of customisation shown in your work. SimpleNet demonstrates similar efficiency to our 1D CNN's results, but in the context of anomalies, our work excels in real-time efficiency within the ICS domain. Without more optimization, TSMAE and AutoML might not be able to match this real-time capability. In terms of applicability to ICS, our work outperforms TSMAE, SimpleNet, and AutoML thanks to its thorough, adaptable, and effective approach to anomaly detection. Our multi-model comparison handled different the anomalies efficiently and, it offers a more flexible and reliable solution for ICS communication anomaly detection than other studies that excel in their specific domains.

V. CONCLUSION & DISCUSSION

PLC vulnerabilities present a broad danger, including operational interruption, illegal access, and possible safety risks. According to [14], ensuring the security of PLC systems is not only an auxiliary worry, but a vital requirement in industrial automation and process control. The greater impact of safeguarding these systems goes beyond individual components, influencing the entire integrity and efficiency of industrial operations.

The suggested solution not only addresses significant PLC vulnerabilities, but it also helps to reduce machine downtime, which is an important aspect in improving production efficiency. By reducing downtime, the method enhances operational resilience and maximizes resource usage in industrial settings. However, the actual test of the presented paradigm is its real-world applicability. Implementing and extensively testing this approach on genuine PLC systems will reveal critical insights into its flexibility and ability in the complexity of real-world industrial settings.

A thorough evaluation that considers both network dynamics and PLC functions will provide a more complete picture of the model's effectiveness. This dual assessment technique ensures a full grasp of the model's performance across several operating scenarios, allowing for a strong defensive mechanism against possible threats. Fine-tuning the model based on ongoing input and performance measurements from real-world deployments will be critical for increasing its resilience and dependability. An iterative strategy based on refinement and adaptation is critical for effectively detecting and reducing risks in industrial control systems.

The 1D CNN's combination of accuracy and processing speed makes it an ideal candidate for real-time deployment in ICS environments. Its ability to detect anomalies promptly can minimize disruptions, prevent damage, and enhance overall system reliability. By addressing anomalies in TCP/IP traffic, this framework provides a versatile solution for protecting ICS against various lateral attack scenarios. The focus on simulated lateral attacks ensures that the models are trained for real-world threats. Despite the

presence of ICS solutions on the market, the specific focus on industrial security, particularly regarding PLCs, remains insufficient. The proposed system aims to bridge this gap by concentrating on industrial networks with a dedicated emphasis on PLCs. Pursuing a specialized system tailored to industrial security is a forward-thinking endeavour, with the potential to significantly enhance the resilience of critical infrastructure. Adopting ML-based anomaly detection systems can significantly strengthen the cybersecurity posture of critical infrastructure sectors such as energy, water, and manufacturing. This research contributes to the growing body of knowledge required to secure these vital systems in the era of Industry 4.0.

While ML-based models offer significant benefits for identifying threats in ICS, they also have drawbacks, such as difficulty obtaining labelled data, vulnerability to adversarial attacks, interpretability problems, resource constraints, and adaptation to changing environments. To solve these problems, it is critical to carefully analyse data gathering strategies, implement model explainability approaches, maintain ongoing monitoring and retraining, and maybe merge ML with traditional security measures. Striking a balance between these parameters is essential for successful and dependable intrusion detection in industrial settings.

In conclusion, continual validation and improvement based on real-world insights are required to show the feasibility and efficacy of sophisticated ML approaches for protecting ICS from future threats. This strategy not only reduces present risks, but also creates the groundwork for incorporating cutting-edge technology in the continued quest of industrial cybersecurity excellence.

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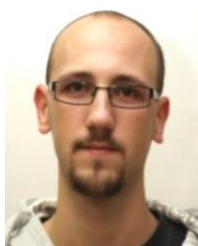
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